Smart Qualitative Data: Methods and Community Tools for Data Mark-Up (**SQUAD**)

Louise Corti and Libby Bishop
UK Data Archive, University of Essex

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Access to qualitative data

- access to qualitative research-based datasets
  - resource discovery points – catalogues
  - online data searching and browsing of multi-media data
  - new publishing forms: re-presentation of research outputs combined with data – a guided tour
- text mining, natural language processing and e-science applications offer richer access to digital data banks
- underpinning these applications is the need for agreed methods, standards and tools

Applications of formats and standards

- standard for data producers to store and publish data in multiple formats
  - e.g UK Data Archive and ESDS Qualidata Online
- data exchange and data sharing across dispersed repositories (c.f. Nesstar)
- import/export functionality for qualitative analysis software (CAQDAS) based on a common interoperable standard
- more precise searching/browsing of archived qualitative data beyond the catalogue record
- researchers and archivists are requesting a standard they can follow – much demand

Our own needs

- ESDS Qualidata online system
  - limited functionality - currently keyword search, KWIC retrieval, and browse of texts
  - wish to extend functionality
    - display of marked-up features (e.g.. named entities)
    - linking between sources (e.g.. text, annotations, analysis, audio etc)

- for 5 years we have been developing a generic descriptive standard and format for data that is customised to social science research and which meets generic needs of varied data types

- some important progress through TEI and Australian collaboration

How useful is textual data?

dob: 1921
Place: Oldham
finalocc: Oldham

[Welham]

U id='1' who='interviewer' Right, it starts with your grandparents. So give me the names and dates of birth of both. Do you remember those sets of grandparents?
U id='2' who='subject' Yes.
U id='3' who='interviewer' Well, we'll start with your mum's parents? Where did they live?
U id='4' who='subject' They lived in Widness, Lancashire.
U id='5' who='interviewer' How do you remember them?
U id='6' who='subject' When we Mum used to take me to see them and me Grandma came to live with us in the end, didn't she?
U id='7' who='Welham' Welham: Yes, when Granddad died - '48.
U id='8' who='interviewer' So he died when he was 48?
U id='9' who='Welham' Welham: No, he was 52. He died in 1948.
U id='10' who='interviewer' But I remember it. How old would I be then?
U id='11' who='Welham' Welham: Oh, you would have been little then.
U id='12' who='subject' I remember him, he used to have whiskers. He used to put me on his knee and give me a kiss.
...

What are we interested in finding in data?

- short term:
  - how can we exploit the contents of our data?
  - how can data be shared?
  - what is currently useful to mark-up?
- long term
  - what might be useful in the future?
  - who might want to use your data?
  - how might the data be linked to other data sets?

What features do we need to mark-up and why?

spoken interview texts provide the clearest—and most common—example of the kinds of encoding features needed

- 3 basic groups of structural features
  - utterance, specific turn taker, defining idiosyncrasies in transcription
  - links to analytic annotation and other data types (e.g.. thematic codes, concepts, audio or video links, researcher annotations)
  - identifying information such as real names, company names, place names, occupations, temporal information

Identifying elements

- Identify atomic elements of information in text
  - Person names
  - Company/Organisation names
  - Locations
  - Dates
  - Times
  - Percentages
  - Occupations
  - Monetary amounts

- Example:
  - Italy's business world was rocked by the announcement last Thursday that Mr. Verdi would leave his job as vice-president of Music Masters of Milan, Inc to become operations director of Arthur Anderson.

How do we annotate our data?

- human effort?
  - how long does one document take to mark up?
    - how much data do you want/need?
    - how many annotators do you have?
  - how well does a person do this job?
    - accuracy
      - novice/expert in subject area
    - boredom
    - subjective opinions
  - what if we decide to add more categories for mark-up at a later date?
- can we automate this?
  - the short answer: "it depends"
  - the long answer...

Automating content extraction using rules

- why don't we just write rules?
  - persons:
    - lists of common names, useful to a point
    - lists of pronouns (I, he, she, me, my, they, them, etc)
      - “me mum”; “them cats”, but which entities do pronouns refer to?

- rules regarding typical surface cues:
  - CapitalisedWord
    - probably a name of some sort e.g. “John found it interesting…”
    - first word of sentences is useless though e.g. “Italy’s business world…
  - title CapitalisedWord
    - probably a person name, e.g. “Mr. Smith” or “Mr. Average”

- how well does this work?
  - not too bad, but…
  - requires several months for a person to write these rules
    - each new domain/entity type requires more time
    - requires experienced experts (linguists, biologists, etc.)

What about more intelligent content extraction mechanisms?

- machine learning
  - manually annotate texts with entities
  - 100,000 words can be done in 1-3 days depending on experience
  - the more data you have, the higher the accuracy
    - the less annotated data you have, the poorer the results
  - if the system hasn't seen it or hasn't seen anything that looks like it, then it can't tell what it is
  - garbage in, garbage out

State of the Art

- use a mixture of rules and machine learning
- use other sources (e.g.. the web) to find out if something is an entity
  - number of hits indicates likelihood something is true
  - e.g.. finding if Capitalised Word X is a country
    - search google for:
      - "Country X"; "The prime minister of X"
- uew focus on relation and event extraction
  - Mike Johnson is now head of the department of computing. Today he announced new funding opportunities.
    - person(Mike-Johnson)
    - head-of(the-department-of-computing, Mike-Johnson)
    - announced(Mike-Johnson, new funding opportunities, today)

Mozilla Firefox

File Edit View Go Bookmarks Tools Help

U id='1' who='interviewer' Right, it starts with your grandparents. So give me the names and dates of birth of both. Do you remember those sets of grandparents?

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U id='12' who='subject' I remember him, he used to have whiskers. He used to put me on his knee and give me a kiss.

U id='13' who='Welham' Welham: Well, he'd have whiskers where he probably had not had a shave and he used to rub her cheek - literally.

U id='14' who='interviewer' What was his occupation?

U id='15' who='Welham' Welham: He worked at ICI, Widness.

U id='16' who='interviewer' Doing what?

U id='17' who='Welham' Welham: What did he do at ICI? Worked at ICI.

U id='18' who='interviewer' What was Grandmother's occupation?

U id='19' who='Welham' Welham: She had nought, because she had arthritis. She couldn't walk. That's why I used to go home every weekend to look after her. We had a Home Help until Friday and I used to go Saturday and stop until Monday when the Home Help came again.

U id='20' who='interviewer' Really? This was when you were married?

U id='21' who='Welham' Welham: Yes. I took me children with me. Me husband, give him his due, he let me go every Saturday, tea-time, until Monday and I used to take the children with me.

U id='22' who='interviewer' And then she came to live with you eventually?

U id='23' who='Welham' Welham: Well, me Dad died. Me sister was younger than me and she got TB and she went in Rufford Sanatorium but I still come every week and looked after me Mum on a Saturday night. Got up, got the dinner ready and got the bus from Widness to Rufford every Sunday to see me sister. Well, she was in for two years in Rufford and then they said they couldn't do nothing, so I brought her home and I nursed her for six weeks and she died and that's when we Mum came to me and we give the house up. There was nothing left. I got three brothers, but wives are not the same as their own daughter. So we all fell out over who was going to have her. So I said well, nobody's having her only me. I brought her to Altrincham and she lived two years. I made her comfortable in the kitchen with a single bed and looked after her. Gave me little job up and I looked after her and she just lived two years.

Key: organisation person location date time money percent geo-political vehicle facility weapon

Done | Radar: | Now: Partly Cloudy, 8° C | Fri: 3° C | Sat: 7° C

UK Data Archive - NLP collaboration

- ESDS Qualidata making use of options for semi-automated mark-up of some components of its data collections using natural language processing and information extraction
- new partnerships created – new methods, tools and jargon to learn!
- new area of application for NLP to social science data
- growing interest in UK in applying NLP and text mining to social science texts – data and research outputs such as publications' abstracts

SQUAD Project: Smart Qualitative Data

Primary aim:

- to explore methodological and technical solutions for exposing digital qualitative data to make them fully shareable and exploitable
- collaboration between
  - UK Data Archive, University of Essex (lead partner)
  - Language Technology Group, Human Communication Research Centre, School of Informatics, University of Edinburgh
- 18 months duration, 1 March 2005 – 31 August 2006

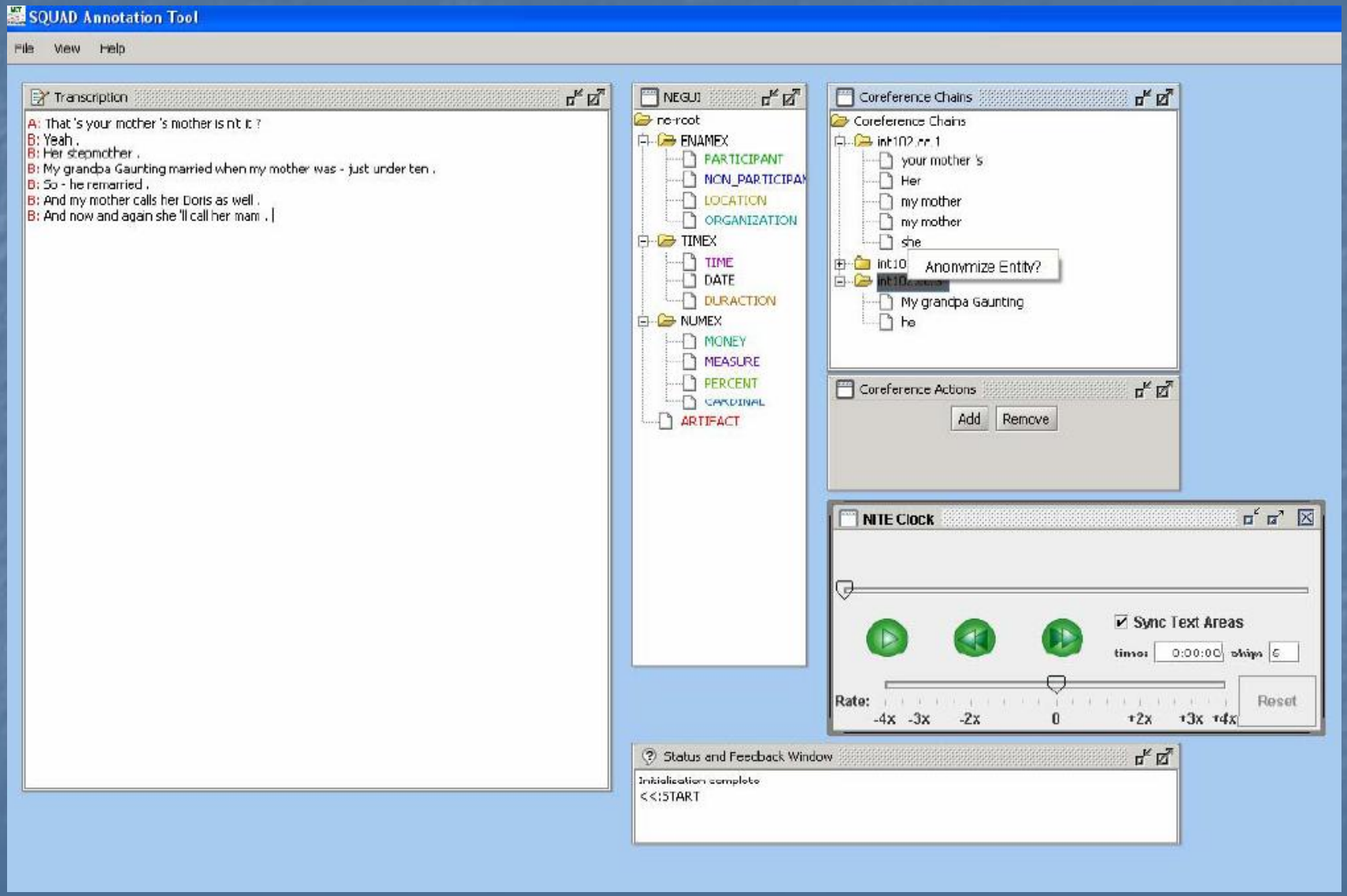
SQUAD: main objectives

- developing and testing universal standards and technologies
  - long-term digital archiving
  - publishing
  - data exchange
- user-friendly tools for semi-automating processes already used to prepare qualitative data and materials (Qualitative Data Mark-up Tools (QDMT)
  - formatted text documents ready for output
  - mark-up of structural features of textual data
  - annotation and anonymisation tool
  - automated coding/indexing linked to a domain ontology
- defining context for research data (e.g.. interview settings and dynamics and micro/macro factors
- providing demonstrators and guidance

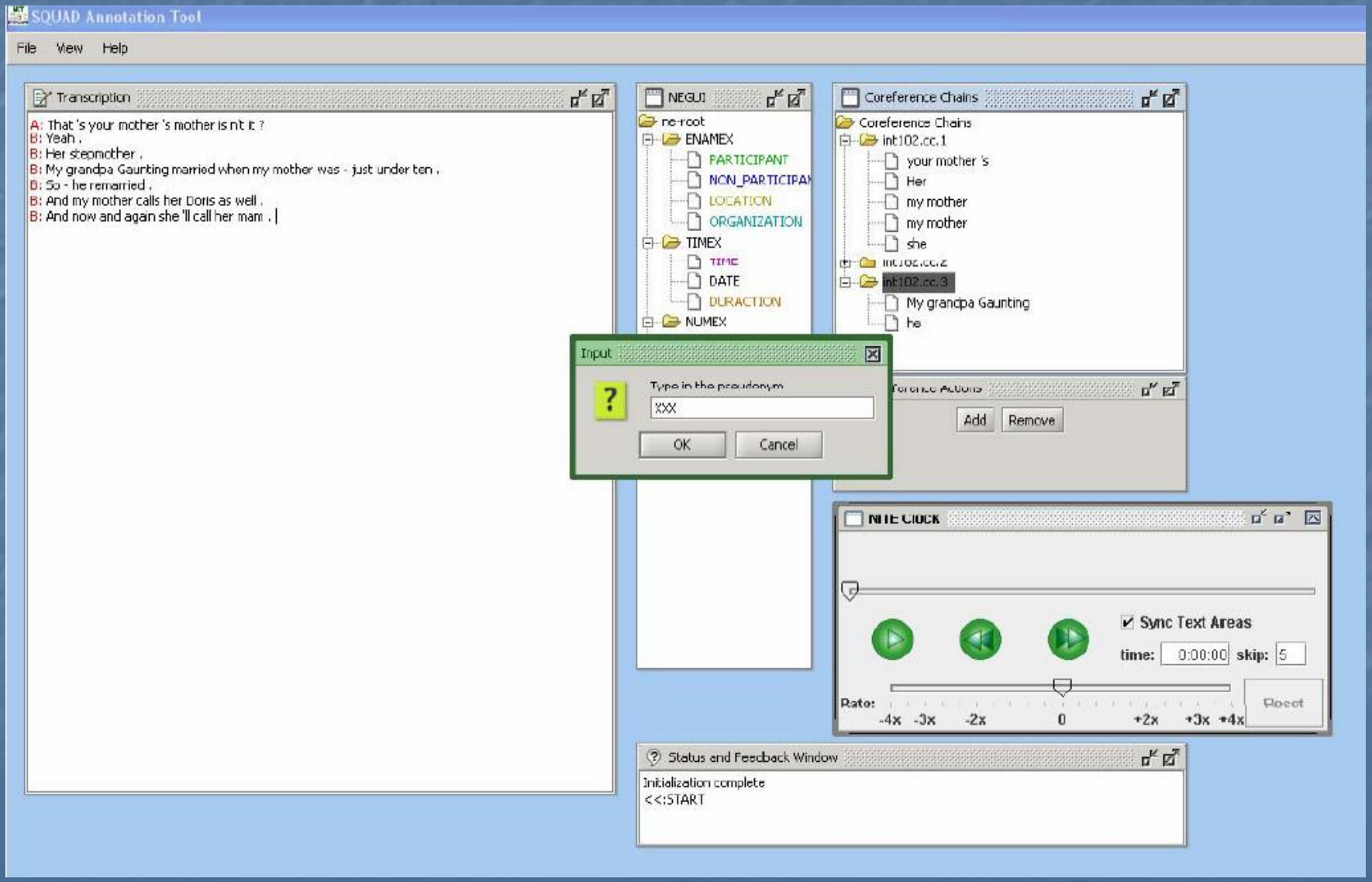
Progress

- draft schema with mandatory elements
- chosen an existing NLP annotation tool - NITE XML Toolkit
- building a GUI – with step-by-step components for 'data processing'
  - data clean up tool
  - named entity and annotation mark-up tool
  - anonymise tool
  - archiving tool – annotated data
  - publishing tool – transformation scripts for ESDS Qualidata Online
- extending functionality of ESDS Qualidata Online system to include audio-visual material and linking to research outputs and mapping system
- from summer:
  - key word extraction systems to help conceptually index qualitative data – text mining collaboration
  - exploring grid-enabling data – e-science collaboration

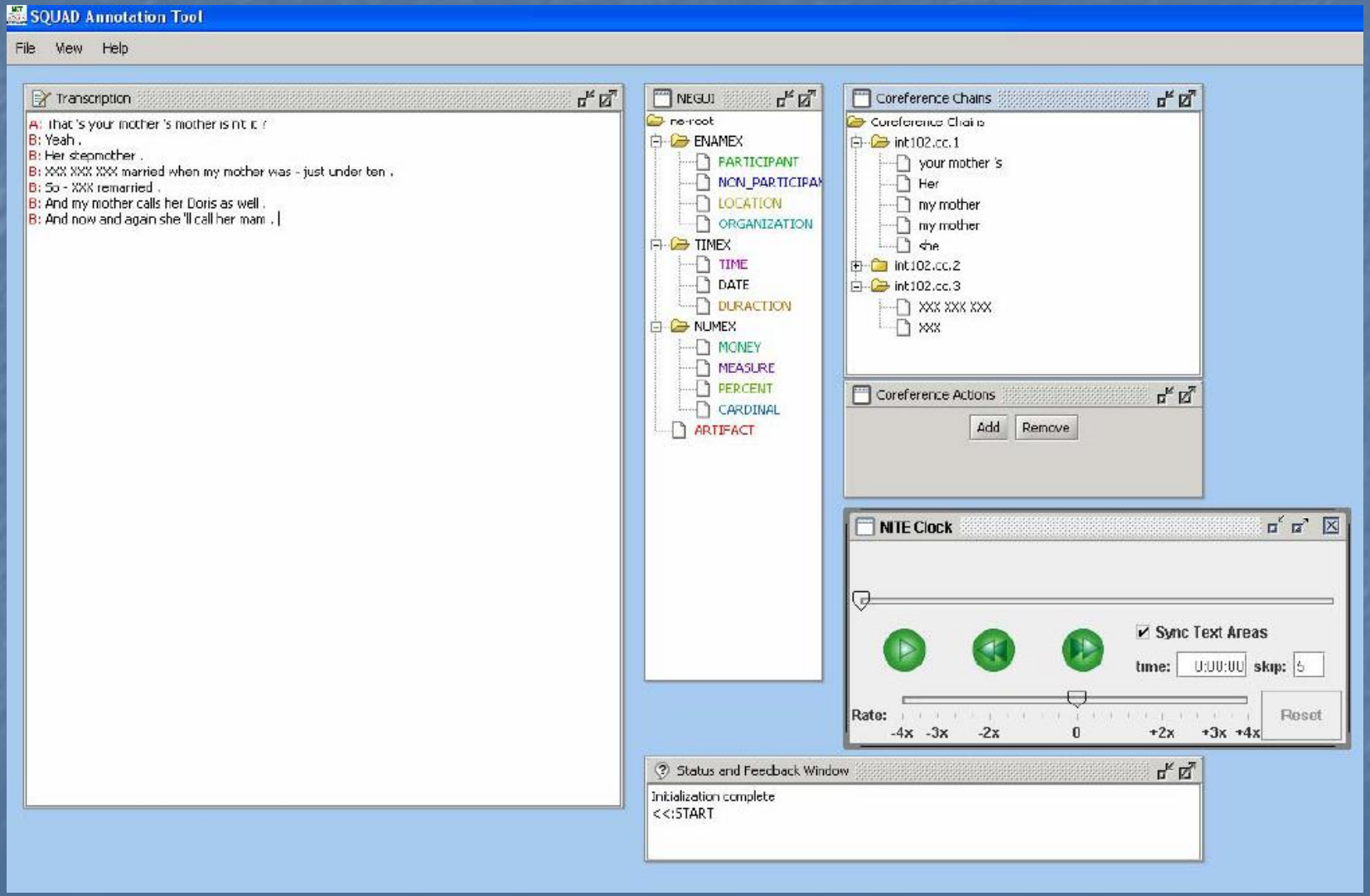
Annotation tool - anonymise



Annotation tool



Anonymised data



Formats - how stored?

- saves original file
- creates new anonymised version
- saved matrix of references - names to pseudonyms
- outputs annotations – who worked on the file etc?
- NITE NXT XML model

- uses ‘stand off’ annotation – annotation linked to or references words

- also about to test Qualitative Data Exchange Format – ANU in Australia
  - non-proprietary exchangeable bundle - metadata, data and annotation
  - testing import and export from Atlas-ti and Nvivo
  - will probably be RDF

Metadata standards in use

- DDI for Study description, Data file description, Other study related materials, links to variable description for quantified parts (variables)
- for data content and data annotation: the Text Encoding Initiative
  - standard for text mark-up in humanities and social sciences
- using consultant to help text the DTD
- will then attempt to meld with the QDIF

ESDS Qualidata XML Schema

"Reduced" set of TEI elements

- core tag set for transcription; editorial changes <unclear>
- names, numbers, dates <name>
- links and cross references <ref>
- notes and annotations <note>
- text structure
- unique to spoken texts <kinesic>
- linking, segmentation and alignment <anchor>
- advanced pointing - XPointer framework
- Synchronisation
- contextual information (participants, setting, text)

Microsoft Excel - workcopy.xls

A22 6.5 Names, numbers, dates+

	A	B	C	D	E
1	DRAFT DTD FOR QUALITATIVE DATA				
2	Partial List of TEI elements for use in ESDS Qualidata XML DTD for transcibed interviews and other qualitative research materials				
3	Citation information: Draft DTD for qualitative data. ESDS Qualidata. Colchester, Essex: UK Data Archive. March 2005.				
4	Revised: 28 June 05 lb				
5	**Ch. Of TEI Guidelines (p5)**	**Element Name**	**Typical Use**	**Attribute**	**Typical use or other comments**
16	**6.3 Highlight and quotation**	<q>	quotation		
17		<emph>	marks words or phrases which are stressed or emphasized for linguistic or rhetorical effect.		
18	**6.4 Editorial changes**				
19		<add>	text added to the document by author, transcriber, etc	place	indicates where the addition has been added by author, scribe, etc. (inline, left[margin], etc.); useful for "end of side n" notations
20				resp	person responsible for addition
21		<unclear>	any text which cannot be transcribed with certainty	reason	illegible, inaudible source, and so on
22	**6.5 Names, numbers, dates+**				
23		<rs>	contains a general purpose name or referring string.		
24		<name>	contains a proper noun or noun phrase		this can take attribute to handle subcategories: person, place, street, address, etc.
25		<num>	any number		
26	**6.6 Links and cross references**				
27		<ref>	reference to another location in the current document	target	can use URI to point to other docs locally or on other systems
28				type	
29		<ptr>	pointer to another location in the current document	targType	
30	**10.2 Elements unique to spoken texts**				

Sheet1

Metadata for model transcript output

Study Name	<titlStmt><titl>Mothers and daughters</titl></titlStmt>
Depositor	<distStmt><depositr>Mildred Blaxter</depositr></distStmt>
Interview number	<intNum>4943int01</intNum>
Date of interview	<intDate>3 May 1979</intDate>
Interview ID	<persName>g24</persName>
Date of birth	<birth>1930</birth>
Gender	<gender>Female</gender>
Occupation	<occupation>pharmacy assistant</occupation>
Geo region	<geoRegion>Scotland</geoRegion>
Marital status	<marStat>Married</marStat>

Transcript with recommended XML mark-up

```
<xml>
<titlStmt><titl>Mothers and Daughters</titl></titlStmt>
<distStmt><depositr>Mildred Blaxter</depositr></distStmt>
<intNum>4943int021</intNum>
<intDate>3 May 1979</intDate>
<persName>g24</persName>
<birth>1930</birth>
<gender>Female</gender>
<occupation>Pharmacy assistant</occupation>
<geoRegion>Scotland</geoRegion>
<marStat>Married</marStat>
<!--comment -->
<u id="1"  who="interviewer">   There's just one or two factual things first of all do you mind my asking how old you are?</u>

<u id="2"  who="subject">       49.</u>

<u id="3"  who="interviewer">   And what schools did you go to?</u>

<u id="4"  who="subject">       <orgName>King Street</orgName>, <orgName>Woodside</orgName> and <orgName>Hilton</orgName>.</u>

<u id="5"  who="interviewer">   Uh-huh .. and how old were you when you left the school?</u>

<u id="6"  who="subject">       14.</u>

<u id="7"  who="interviewer">   And you work at the moment? What sort of work do you do?</u>

<u id="8"  who="subject">       Well I've gone back to get shorter hours, I've went back to domestic, which I dinna really
care for. But then I used to be in the pharmacy department at <orgName>ARI</orgName> ... just <occupation>pharmacy
assistant</occupation>. At least it was better than cleanin'! But then they've nae part-time workers there so..</u>

<u id="9"  who="interviewer">   And did you work in the pharmacy long?</u>

<u id="10"  who="subject">      I was there for eleven years.</u>

<u id="11"  who="interviewer">  And did you have any other sort of jobs?</u>

<u id="12"  who="subject">      Where? Since I left school, like? Well, when I first left school I was just a <occupation>shop
 assistant</occupation> in a number of shops like <orgName>Reid</orgName> and <orgName>Pearsons</orgName>, which is.. we hinna
got it ony mair.</u>

<u id="13"  who="interviewer">  Gone ..!</u>

<u id="14"  who="subject">      <orgName>Marks and Spencers</orgName>, and the <orgName>Coop</orgName>. And that's aboot
it.</u>

<u id="15"  who="interviewer">  Uh-huh .. and then you got married and had your family. And when you went back to work, did
you go into the pharmacy?</u>
```

XML is source for .rtf download

Study Name:: Mothers and Daughters
Depositor: Mildred Blaxter
Interviewer: Liz Patterson

Interview number: 4943int021.doc
Interviewer ID: g24
Date of interview: 3 May 1979

Information about interviewee
Date of birth: 1930
Gender: female
Marital status: married
Occupation: pharmacy assistant
Geographic region: Scotland

LP: There's just one or two factual things first of all do you mind my asking how old you are?

G24: 49.

LP: And what schools did you go to?

G24: King Street, Woodside and Hilton.

LP: Uh-huh .. and how old were you when you left the school?

Metadata used to display search results

ID: g24
Born: 1930 Female
Occupational Class: Sales and customer service **Geographic Region:** Scotland
... and Hilton. Uh-huh. and how old were you when you left the **school**? 14. And you work at the moment? What sort of work do you **...**

ID: g25
Born: 1932 Female
Occupational Class: Elementary **Geographic Region:** Scotland
... No, no, no. I was a shop assistant when I left **school**. And then we had our own shop down in Ashgrove there. my sister an **...**

ID: g31
Born: 1929 Female
Occupational Class: Unemployed **Geographic Region:** Scotland
... , what sort of work did you do? When I first left the **school**? An upholstery sewer. I worked in Roberts in Union Street for 16 years, **...**

ID: g32
Born: 1921 Female
Occupational Class: Unemployed **Geographic Region:** Scotland
... Gordon's College. And a shop assistant before that. And when you left **school**? Oh, when I left **school** I went to Broadford. When I left **school** **...**

ID: g6
Born: 1921 Female
Occupational Class: Elementary **Geographic Region:** Scotland
... . I'm 57. And what schools did you go to? Old Aberdeen **School** for a start, and then Sunnybank. And how old were you when you left **...**

ID: g7
Born: 1916 Female
Occupational Class: Unemployed **Geographic Region:** Scotland
... what schools did you go to yourself? Well, Porthill and the Middle **School**. Aye, Porthill's the Primary and the Middle's the Secondary. Yes. And how **...**

You are here : ESDS Qualidata Online > Explore > IndexZoom > Interview transcripts: search results

XML+XSL enables online publishing

ESDS Qualidata*Online*

esds
Economic and Social Data Service

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Introduction

Edwardians

Mothers and Daughters (SN 4943)

100 Families (SN 4938)

Transcript

To view the interview summary, click the title. (Only available for Edwardians interviews at the moment).

g24

There's just one or two factual things first of all do you mind my asking how old you are?

49.

And what schools did you go to?

King Street, Woodside and Hilton.

Uh-huh .. and how old were you when you left the school?

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And did you work in the pharmacy long?

I was there for eleven years.

And did you have any other sort of jobs?

Information

- ESDS Qualidata Online site:
  www.esds.ac.uk/qualidata/online/
- SQUAD website:
  quads.esds.ac.uk/projects/squad.asp
- NITE NXT toolkit:
  www.ltg.ed.ac.uk/NITE
- ESDS Qualidata site:
  www.esds.ac.uk/qualidata/

We would like collaboration and testers!